Neural Genome Processing and Network Visualization

Overview

This system models a simple feedforward neural structure with the following architecture:

- $N_s = 8$ sensory (input) neurons
- $N_h = 4$ internal (hidden) neurons
- $N_a = 5$ action (output) neurons
- $N_c = 10$ connections encoded as 32-bit genes

Gene Structure

Each gene is a 32-bit binary string (represented as 8 hex characters) encoding a single neural connection. The structure is:

$$\underbrace{b_0}_{\text{source\_type}} \underbrace{b_{1-7}}_{\text{source\_id}} \underbrace{b_8}_{\text{sink\_type}} \underbrace{b_{9-15}}_{\text{sink\_id}} \underbrace{b_{16-31}}_{\text{weight}}$$

- **source_type**: 0 for sensory neuron, 1 for internal neuron
- **sink_type**: 0 for internal neuron, 1 for action neuron
- **weight**: 16-bit signed integer mapped to real value in $[-4.0, 4.0]$

Activation Propagation

The network computes values as follows:

1. Propagate values from sensory neurons to internal or action neurons.
2. Store connections from internal $\rightarrow$ internal and internal $\rightarrow$ action for delayed execution.
3. Evaluate internal $\rightarrow$ internal, then apply tanh activation to internal layer.

4. Evaluate internal → action, then apply tanh to action layer.

Finally, the softmax of the action layer is used to probabilistically select an action.

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

Example Brain Structure

An example graph of the neural architecture from a randomly generated genome is shown below. Connections are color-coded:

- Blue: excitatory ($w > 0$)
- Red: inhibitory ($w < 0$)

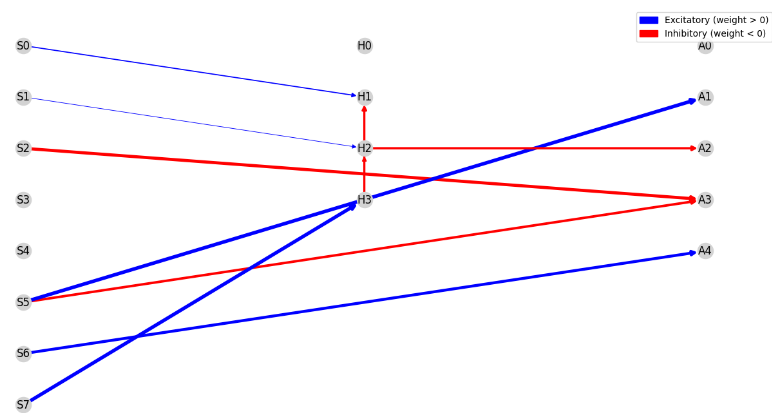

Figure 1: Graphical representation of the neural genome structure